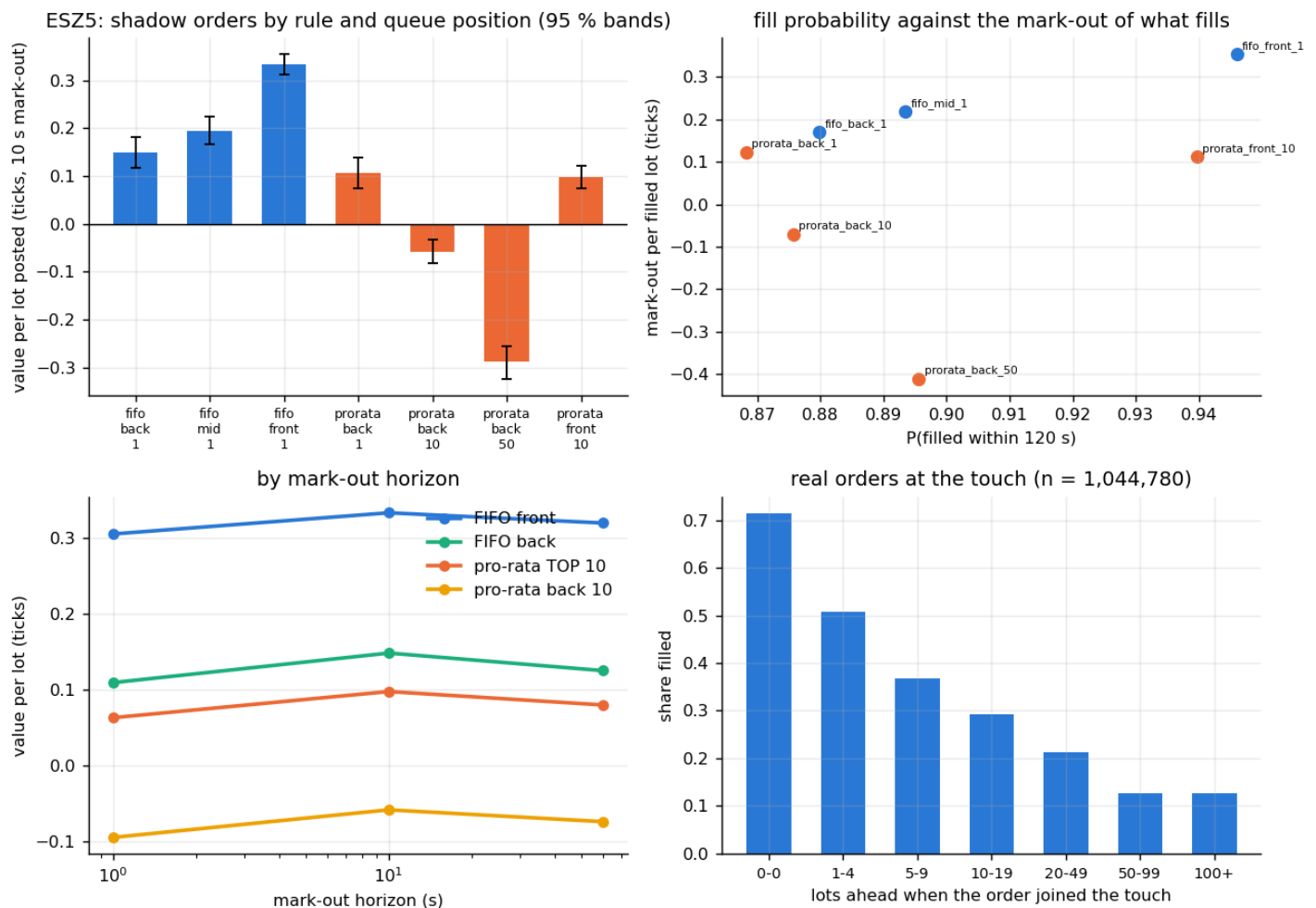


Futures Desk Microstructure and Analytics on CME

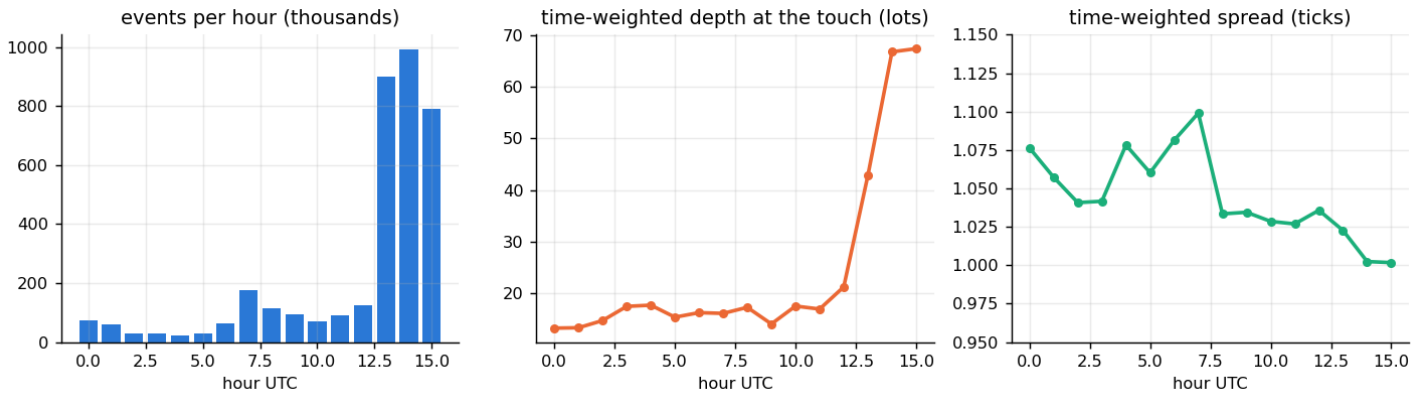
Question. A futures desk on CME lives on four mechanics: the exchange's matching rules (FIFO for the index, Treasury and energy outright; pro-rata with a TOP order for SOFR), the queue position of its passive orders, the quarterly or monthly roll of every position it carries, and a P&L that only makes sense once daily settlement variation, the roll, fees and the financing of margin are separated. What is a place in the queue worth under each matching rule, measured on recorded order-by-order depth rather than assumed; how do implied orders move an outright book; what does a roll cost per contract and how does the spread behave in the roll window; how does a stated futures book's P&L decompose; and how fast does a positions-and-limits monitor catch what goes wrong?

Method. Python package (futdesk) with a C++ replay (pybind11) and DuckDB/parquet storage. Matching engine: price-time, the Allocation algorithm (TOP order, pro-rata with a 2-lot minimum, FIFO leftover), the configurable FIFO/pro-rata split, and first-generation implied orders between two outright and their calendar spread, with known-answer tests. Book replay: the order book rebuilt from every MBO event and checked level by level against the exchange's MBP-10 feed. Queue value: shadow orders inserted at the touch every second at the front, middle and back of the queue under FIFO and under pro-rata, carried through the recorded fills and cancels, valued at the mark-out of what fills, with block-bootstrap bands; and the outcome of every real order posted at the touch. Rolls: calendar spreads between listed contracts from daily closes, the implied financing of the equity-index spread against SOFR net of dividends, the spread and the volume migration through the roll window, and forty years of monthly energy rolls from the EIA settlement history. Attribution: the stated book on settlements (price, roll, execution, financing; core and hedge), and a passive one-lot quoter replayed on the tape (spread capture, inventory, hedge, fees), both reconciling to the cent. Monitor: thirteen rules, eleven fault kinds injected on simulated days built on real price paths, detection rate and false alerts scored.

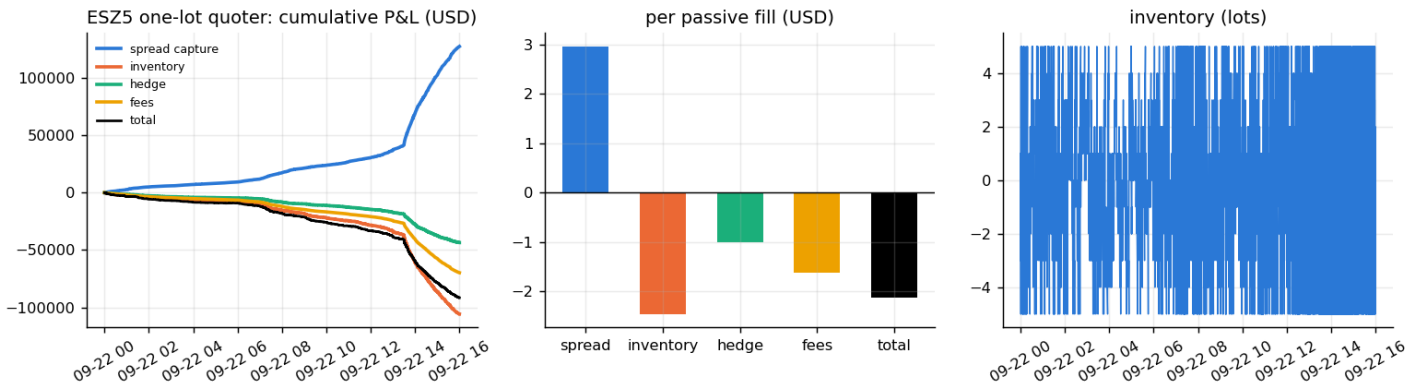
What is real and what is simulated. The order-by-order tape is real (Databento's public CME Globex MDP 3.0 sample: one Globex session of the December 2025 E-mini S&P 500, 3.7 million events); the settlements are real daily closes per listed contract (Yahoo) and EIA's NYMEX settlement history; SOFR is the NY Fed's. The shadow orders, the quoter and the pro-rata counterfactual are simulations on that tape and assume our order would not have changed anyone else's behaviour. The monitor's days are simulated. Fees and margins are approximate public values in a config file, to be replaced by the desk's own.



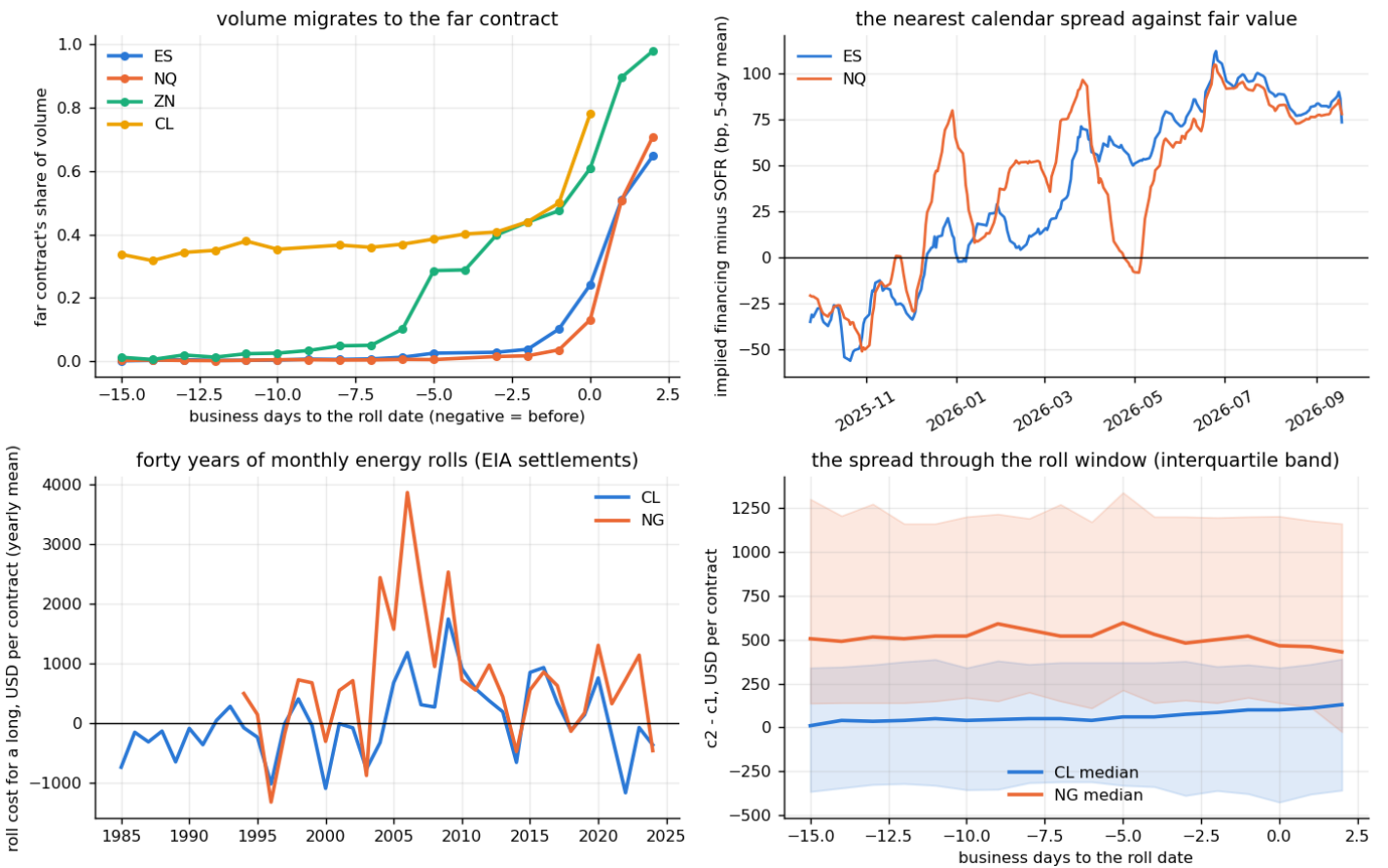
Queue position on the recorded ESZ5 tape: value per lot posted by rule and place in the queue with bootstrap bands; fill probability against the mark-out of what fills; the value by mark-out horizon; and the fill share of every real order by the volume ahead of it when it joined the touch.



The ESZ5 book through the session: events per hour, time-weighted depth at the touch and the time-weighted spread (one tick 95 % of the time).



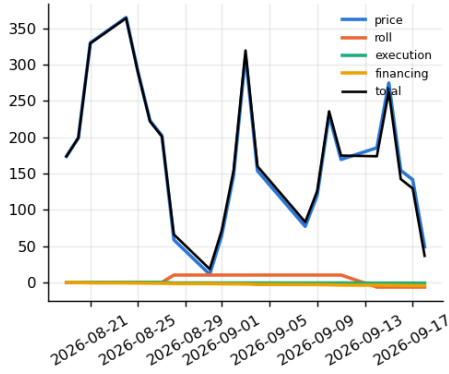
The passive one-lot quoter replayed on the tape: cumulative spread capture, inventory, hedge and fees; the per-fill decomposition; the inventory path.



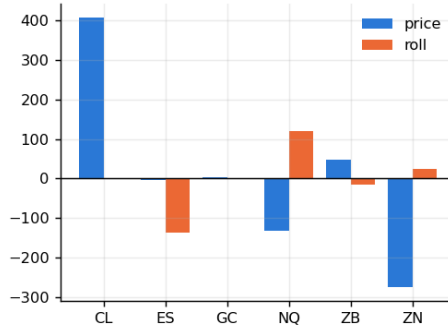
Rolls: the far contract's share of volume through the roll window; the equity-index calendar spread's implied financing against SOFR net of dividends; forty years of monthly energy roll costs; the CL and NG spread through the roll window.

futures-desk-analytics - Globex matching, queue value on recorded depth, rolls, P&L attribution, the monitor

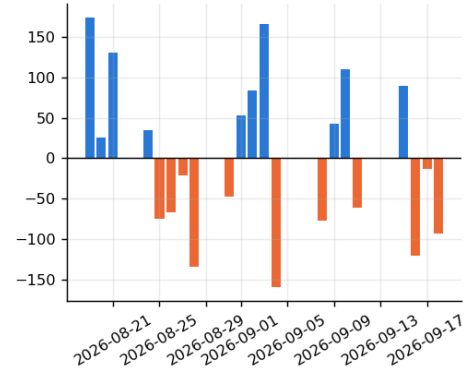
the stated book: cumulative components (USD thousands)



by product (USD thousands)

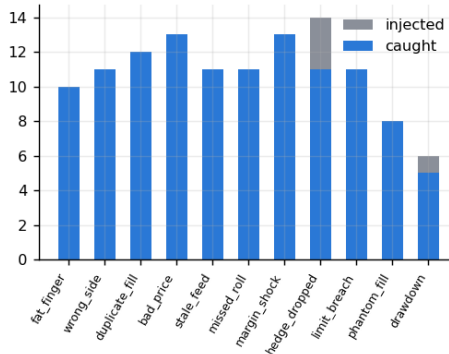


daily settlement variation (USD thousands; margin 1.4)

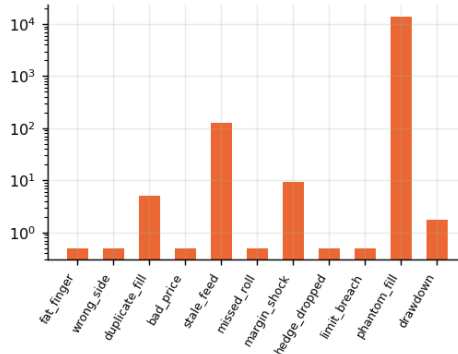


The stated book on settlements: cumulative price, roll, execution and financing components; by product; the daily settlement variation.

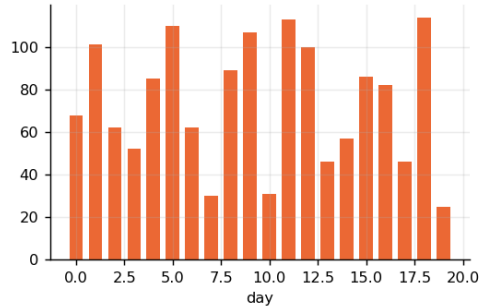
faults by kind



median time to detect (s, log)

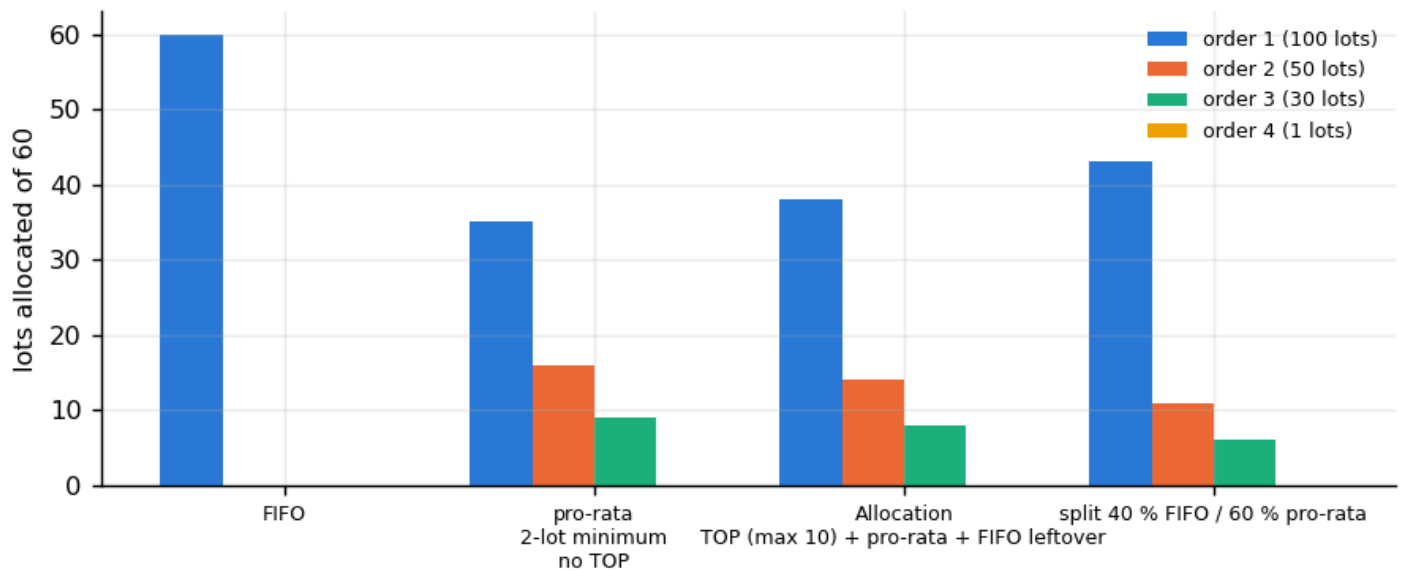


alerts per day (orange = fault days, aqua = clean)



The monitor: faults injected and caught by kind, the median time to detect, alerts per day on fault and clean days.

the allocation rules on one level



The allocation rules on one price level: FIFO, pro-rata with a 2-lot minimum, the Allocation algorithm with a TOP order, and the 40/60 split.

Generated 2026-09-19T07:10:25+00:00 in 61 s (full run; compiled replay: True).

Tapes

file	symbol	events	session (UTC)	hours	book rebuilt vs MBP-10
mbo_glbx-mdp3-sample.parquet	ESZ5	3,667,804	2025-09-22T00:00 -> 15:59	16.0	67 of 147,275,160 fields differ over 2,45

ESZ5: the book

3,658,972 events, 1,430,012 orders added, 153,819 trades for 493,504 lots (mean 3.2, p99 28); time-weighted spread 1.045 ticks (one tick wide 95.5% of the time), depth at the touch 24.3 lots in 15.3 orders; exchange matching algorithm F.

						hour UTC
00	72,810	3,375	1.076	13.3	10.0	
01	59,514	2,738	1.057	13.3	9.8	
02	31,022	1,345	1.041	14.7	9.9	
03	30,885	1,433	1.042	17.5	12.1	
04	20,841	1,067	1.078	17.7	11.3	
05	29,761	1,393	1.060	15.4	10.2	
06	61,822	3,220	1.081	16.2	11.2	
07	176,401	7,147	1.099	16.1	10.9	
08	114,844	4,975	1.033	17.3	11.7	
09	93,209	3,516	1.034	14.1	10.3	
10	71,178	3,265	1.029	17.5	11.9	
11	90,140	4,305	1.027	16.9	11.1	
12	123,473	6,518	1.036	21.2	14.0	
13	899,842	37,636	1.023	42.8	25.8	
14	992,944	41,029	1.002	66.7	37.3	
15	790,286	30,857	1.002	67.4	36.8	

ESZ5: queue-position value (shadow orders, 10 s mark-out)

57,540 insertion times, 402,780 shadow orders, 1,836,867 fills; 192 five-minute bootstrap blocks, 1000 draws.

										rule
fifo	back	1	0.880	0.880	9.3	0.313	0.169	0.149	[0.118, 0.181]	
fifo	mid	1	0.893	0.893	7.1	0.391	0.218	0.195	[0.166, 0.225]	
fifo	front	1	0.946	0.946	3.4	0.501	0.353	0.334	[0.313, 0.355]	
prorata	back	1	0.868	0.868	9.7	0.278	0.123	0.107	[0.074, 0.138]	
prorata	back	10	0.876	0.825	14.5	0.105	-0.070	-0.058	[-0.082, -0.032]	
prorata	back	50	0.896	0.701	29.0	-0.244	-0.411	-0.288	[-0.323, -0.255]	
prorata	front	10	0.940	0.877	8.9	0.298	0.112	0.098	[0.074, 0.122]	

FIFO: front of the queue minus the back (one lot): 0.185 ticks per lot [0.164, 0.206] = \$2.31 [2.05, 2.57]

FIFO: middle minus the back: 0.046 ticks per lot [0.039, 0.053] = \$0.58 [0.48, 0.67]

pro-rata: TOP minus the back (ten lots): 0.156 ticks per lot [0.145, 0.167] = \$1.95 [1.81, 2.08]

pro-rata: fifty lots minus one lot, per lot: -0.394 ticks per lot [-0.430, -0.361] = \$-4.93 [-5.37, -4.52]

back of the queue: FIFO minus pro-rata (one lot): 0.042 ticks per lot [0.034, 0.052] = \$0.53 [0.42, 0.65]

FIFO front minus back, rth only (63,000 shadows): 0.055 ticks per lot [-0.005, 0.105] = \$0.68 [-0.06, 1.31]

FIFO front minus back, overnight only (339,780 shadows): 0.209 ticks per lot [0.188, 0.231] = \$2.61 [2.35, 2.88]

FIFO front minus back at a 1 s mark-out: 0.196 [0.178, 0.214]

FIFO front minus back at a 60 s mark-out: 0.195 [0.171, 0.221]

ESZ5: every real order posted at the touch

1,044,780 orders (mean size 1.82, 66% one lot): 26.9% filled, 72.8% cancelled; median life 0.17 s (filled 0.49 s, cancelled 0.12 s).

						lots ahead
0-0	42,980	0.714	0.00	2.16	0.037	
1-4	86,247	0.507	0.02	1.84	0.042	
5-9	115,888	0.367	0.17	1.73	0.044	
10-19	191,908	0.292	0.22	1.79	0.043	
20-49	363,514	0.213	0.14	1.80	0.022	
50-99	224,627	0.126	0.31	1.83	0.019	

100+	19,616	0.128	0.43	1.90	0.021
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ESZ5: the passive one-lot quoter on the tape

42,946 passive fills and 1026 hedges (5130 lots) over 16.0 h; inventory limit 5, max |inventory| 5.

spread_capture	127,369	2.97
inventory	-105,694	-2.46
hedge	-43,500	-1.01
fees	-69,710	-1.62
total_after_fees	-91,535	-2.13

Identity check (cash + inventory at the closing mid - spread capture - inventory - hedge): 0.00e+00 ticks.

Rolls

product	contracts	pair-days	rolls observed	roll cost for a long, USD/contract	spread the prior 10 bd (USD)	implied financing - SOFR, last year
ES	5	2,000	1	3,392	-12	51 [43, 58]
NQ	5	1,005	1	5,948	25	39 [33, 46]
ZN	3	190	1	-244	16	-
ZB	3	190	1	-490	0	-
ZT	3	176	1	-268	-47	-
SR3	13	12	0	-	-	-
ZQ	18	8,518	0	-	-	-
CL	35	17,035	1	-4,212	-1,310	-
NG	36	17,539	0	-	-	-
GC	14	3,498	0	-	-	-

product	event	roll date	spread	USD/contract	drift 10 bd (USD)	volume crossover (bd b	implied financing - SO
ES	ESU26->ESZ26	2026-09-11	67.7500	3,388	-12	-1	91
NQ	NQU26->NQZ26	2026-09-11	297.2500	5,945	25	-1	86
ZN	ZNU26->ZNZ26	2026-08-27	-0.2500	-250	16	0	-
ZB	ZBU26->ZBZ26	2026-08-27	-0.5000	-500	0	0	-
ZT	ZTU26->ZTZ26	2026-08-27	-0.1367	-273	-47	0	-
CL	CLV26->CLX26	2026-09-18	-4.2200	-4,220	-1,310	0	-

Energy rolls from the EIA settlement history

product	rolls	span	mean roll cost for a l	95 %	mean absolute	share in contango	drift over the prior 1
CL	471	1985-01 -> 2024-03	38	[-48, 124]	595	57%	-17 [-60, 28]
NG	363	1994-01 -> 2024-03	726	[519, 942]	1,244	80%	-205 [-340, -81]

The stated book

2026-08-19 -> 2026-09-18 (22 settlement days; average notional \$48m, initial margin \$1.45m). Positions: +40 ES (core), -20 NQ (hedge), +100 ZN (core), -30 ZB (hedge), +25 CL (core), +10 GC (core).

price	48,713
roll	-6,600
variation	42,113
execution	-1,080
financing	-4,564
total	36,469

variation = price + roll to 0.00e+00; daily variation sd \$96,773, worst day \$-158,950 (2026-09-04), max drawdown \$-345,563; financing is 0.26% of the gross daily variation.

CL	406,000	406,000	0	0	0
ES	-138,500	-3,000	-135,500	-166	1
GC	2,900	2,900	0	0	0
NQ	-13,600	-132,500	118,900	-68	1
ZB	33,750	48,750	-15,000	-285	1
ZN	-248,438	-273,438	25,000	-561	1

role	variation	price	roll	execution
core	21,963	132,463	-110,500	-727
hedge	20,150	-83,750	103,900	-353

The monitor

20 fault days and 20 clean days (simulated on real price paths): 116 of 120 faults caught (96.7%), median time to detect 0.0 s; 0.00 alerts per clean day, 73.3 per fault day.

					fault
fat_finger	10	10	0.0	FAT_FINGER	
wrong_side	11	11	0.0	POSITION_JUMP	
duplicate_fill	12	12	5.0	DUPLICATE_FILL	
bad_price	13	13	0.0	PRICE_BAND	
stale_feed	11	11	128.5	STALE_MARK	
missed_roll	11	11	0.0	DELIVERY_RISK, EXPIRY_RISK	
margin_shock	13	13	9.5	MARGIN_CALL	
hedge_dropped	14	11	0.0	PAIR_BROKEN	
limit_breach	11	11	0.0	POSITION_LIMIT	
phantom_fill	8	8	13,893.0	RECON_BREAK	
drawdown	6	5	1.7	PNL_DRAWDOWN	

The matching rules on one level

Resting [100, 50, 30, 1] lots (order 1 has TOP status), 60 lots incoming.

					rule
FIFO	60	0	0	0	
pro-rata, 2-lot minimum, no TOP	35	16	9	0	
Allocation: TOP (max 10) + pro-rata + FIFO	38	14	8	0	
split 40 % FIFO / 60 % pro-rata	43	11	6	0	

Implied spread: leg1 1000/1002, leg2 990/993 -> implied spread bid 7 (6 lots), ask 12 (4); selling 2 spreads at 7 executed 2 leg fills and left leg1 bid 8 lots, leg2 ask 4 lots.